EduMatrix

Next-Gen Smart Campus & Academic Intelligence Platform

Official System Architecture, Role-Based Usage Manual & Operational Workflow Guide

EXECUTIVE SUMMARY

EduMatrix is an institutional-grade platform unifying AI Biometrics, Academic Ledgers (GIET ERP Live Sync), Placement Readiness Engines, Smart Fee Portals with Parent SMS alerts, and Campus Life Gamification. Standardized for all B.Tech Batches (2023 - 2027).

1. Role-by-Role Features & Usage Matrix ('What Uses for Whom')

Comprehensive breakdown of modules accessible by Students, Teachers, HODs, Admins, and Parents

ROLE: STUDENT (E.G. B.TECH BATCH 2023-2027)

- **Academic Ledger & GIET ERP:** View 100% synchronized live overall attendance (e.g. 85.0%), subject-wise attendance breakdowns, semester-wise SGPA/CGPA calculations, and official academic records.
- **AI Placement Readiness Suite:** Practice 1-on-1 virtual interviews with Ask EduMatrix AI, build ATS-ready industry resumes (PDF export), and attempt standardized aptitude & technical mock tests.
- **Smart Fee Payment & Parent Request:** View tuition, hostel, exam, and library dues. Pay online via UPI/Card OR send an instant payment request to parents with automatic SMS/WhatsApp notification dispatch to their mobile number.
- **Smart Notices & Circulars:** Instant access to prioritized institution circulars, department announcements, and exam timetables with read-tracking.
- **Campus Life & Gamification:** Order food with live queue status on EduMatrix Bites, study with Focus Flow XP timer, log daily mood on Campus Pulse, and redeem gift store rewards.
- **Smart Library & Spaces:** Reserve library study seats in real time with live seat matrices and book campus study rooms.
- **Class Messaging & Chat:** Group discussions with classmates and 1-on-1 direct messaging with subject faculty.

ROLE: TEACHER / FACULTY MEMBER

- **AI Facial Attendance Scanner:** Capture real-time face embeddings from classroom camera, automatically recognize students, and record live attendance in seconds with anti-spoofing.
- **Real-Time Class Monitoring:** Live class feed showing present, late, and absent students with automated attendance percentage recalculation.
- **Student Academic Lookup:** Search any student by roll number to instantly inspect their attendance status, internal marks, ERP ledger, and contact details.
- **Leave Request Approvals:** Approve or reject student leave requests submitted for their assigned classes or sections.
- **Internal Marks Upload:** Record Midterm 1, Midterm 2, assignment, and lab practical marks individually or via bulk assessment upload.
- **Availability Management:** Toggle active office hours status and update custom availability notes visible to students.

ROLE: HEAD OF DEPARTMENT (HOD)

- **Department-Wide Monitoring:** Analyze overall attendance trends, track high-absence subjects, and identify defaulters across all B.Tech batches (2023-2027).
- **Faculty & Section Leaves:** Review and approve faculty leave applications and section-level requests.
- **Curriculum & Timetable Allocation:** Manage subject teachers, assign mentors/proctors, and schedule makeup classes.

ROLE: SYSTEM ADMINISTRATOR (INSTITUTIONAL ADMIN)

- **Web Administrative Dashboard:** Full web-based management suite for user provisioning (Students, Teachers, Staff), batch assignments, and master timetables.
- **Emergency & Mass Broadcasts:** Send high-priority push notifications and circulars targeting specific departments, years, or

campus-wide.

- Defaulter Management: Automated detection of students with < 75% attendance criteria and one-click notification dispatch to student and guardian.

ROLE: PARENT / GUARDIAN

- Fee Payment Request Alerts: Receive automated SMS/WhatsApp notifications on their phone number with student roll no, fee category, amount due, and direct payment details.

- Attendance & Safety Alerts: Receive immediate alerts when attendance falls below criteria or when critical campus circulars are issued.

2. System Architecture & Technical Workflows ('How It Works')

Deep-dive into the AI recognition pipeline, ERP synchronization, and automated communication engines

A. AI Facial Recognition Engine (Biometric Attendance)

*** 128-Dimensional Deep Metric Face Embeddings**

1. Image Ingestion: Camera frames captured via Android CameraX or Web Cam.
2. Face Detection: Multi-task Cascaded CNN / HOG detects facial bounding boxes.
3. Feature Extraction: dlib ResNet-34 model extracts a 128-dimensional floating point vector per face.
4. Encodings Database: Pre-registered student face encodings are cached in memory on backend startup.
5. Distance Matching: Euclidean distance is computed against known encodings. Matches with distance <= 0.50 are marked 'Present'.
6. Anti-Spoofing & Validation: Validates student section, subject slot, and date-time integrity.

B. GIET ERP & Academic Ledger Synchronization

*** Real-Time Bi-Directional Ledger Alignment**

1. Normalized Student Batch: All B.Tech students are indexed under batch '2023 - 2027'.
2. Dynamic Calculation: Queries total classes conducted vs attended from the Attendance database.
3. Live Synchronization: Ensures Student Dashboard (UI banner) and GIET ERP Screen always render the exact same percentage (e.g. 85.0%).
4. Fallback Resilience: Built-in local caching guarantees zero downtime even during university ERP maintenance.

C. Smart Fee Portal & Parent SMS/WhatsApp Notification Pipeline

*** Automated Multi-Channel Notification Workflow**

1. Dues Assessment: Student opens Fee Portal -> queries live tuition, hostel, exam, and library dues.
2. Student Request Trigger: Student taps 'Request Parent to Pay' -> enters parent name and mobile number.
3. DB Persistence: Saves record into `ParentPaymentRequest` and `Notification` database tables.
4. Firebase Cloud Messaging (FCM): Dispatches instant push notification to student and linked accounts.
5. SMS / WhatsApp Dispatch Gateway: Backend dispatches instant formatted SMS to parent phone number:
   'Dear [Parent], GIET University generated fee payment request for [Student] ([Roll No]) of Rs. [Amount] for [Category]. Please pay via EduMatrix portal.'
6. Strong Typing: Uses typed Retrofit DTOs (FeePayRequestDto, ParentPaymentRequestDto) with @JvmSuppressWildcards for 100% crash-free network operations.

D. AI Placement & Career Readiness Engine

*** Interactive AI Mock Interview & Resume Generator**

1. AI Mock Interviewer: Uses Google Gemini AI / LLM APIs to generate dynamic technical & HR questions tailored to student's branch (CSE/ECE/EE/MECH).
2. Real-Time Feedback: Analyzes user's spoken/text answers and scores clarity, technical accuracy, and areas for improvement.
3. Standardized Test Center: Evaluates quantitative aptitude, logical reasoning, and domain coding skills.
4. Resume Builder: Generates professional single-page ATS-compliant resumes with PDF export capability.

3. Step-by-Step User & Deployment Guide

Complete operating procedures for students, faculty, and developers

Step-by-Step: Student Everyday Operations

- Step 1 (Login): Open EduMatrix app -> Enter Student Username & Password -> Tap Login.

- Step 2 (Check Attendance): View the prominent 'Overall Attendance' gauge on the main dashboard. Tap 'GIET ERP Dashboard' for subject-by-subject attendance.

- Step 3 (Fee Payment / Parent Request): Go to 'Fee Payment' -> Select Fee Category (Tuition/Hostel) -> Tap 'Request Parent to Pay' -> Enter parent's phone number -> Submit. Parent receives SMS instantly.

- Step 4 (Smart Notices): Tap 'Priority Notices' to read university circulars, exam schedules, and holiday announcements.

- Step 5 (Placement Preparation): Navigate to 'Placement Readiness' -> Start 'AI Mock Interview' -> Answer interview questions to receive score and improvement advice.

Step-by-Step: Faculty Attendance Taking & Monitoring

- Step 1 (Login): Login with Teacher / Faculty credentials -> Land on Teacher Dashboard.

- Step 2 (Live AI Attendance): Tap 'Facial Attendance' -> Select Department (CSE), Year (4th Year), Section (G), Subject -> Point camera towards classroom -> Faces are automatically recognized and recorded.

- Step 3 (Student Lookup): Tap 'Student Academic Lookup' -> Search any roll number (e.g. 23CSE706) to inspect academic standing and attendance history.

- Step 4 (Marks Upload): Go to 'Upload Marks' -> Select subject -> Input Midterm/Assignment scores -> Submit.

Step-by-Step: Developer & Server Setup

*** Running Backend & Android Client**

A. Backend Execution (FastAPI Server on Port 8000):

1. Open Terminal: cd "D:\7th Sem\Project\EduMatrix"
2. Run Server: py -3.10 -m uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
3. Web Dashboard: http://localhost:8000/ | Swagger Docs: http://localhost:8000/docs

B. Mobile APK Build & Execution:

1. APK File: D:\7th Sem\Project\app-debug.apk
2. Android Studio: Open 'D:\7th Sem\Project\EduMatrix_App' -> Enable USB Debugging -> Press Run (Shift + F10).
3. Local Network: Connect phone & PC to same Wi-Fi (IP: 10.49.123.225:8000).

Summary Specifications Table

Component	Technology Stack	Key Responsibility
Mobile Client	Kotlin, Jetpack Compose, Hilt, Retrofit	Clean Architecture Android application for all roles
Backend Server	FastAPI (Python 3.10), Uvicorn, Pydantic	High-performance REST API, WebSockets & background tasks
AI Biometrics	dlib, OpenCV, ResNet-34 Face Encodings	Real-time facial detection and biometric recognition
Database Layer	SQLAlchemy ORM, MySQL / SQLite	Relational persistence for users, attendance, fees & marks
Notifications	Firebase Cloud Messaging (FCM), SMS Gateway	Real-time push circulars and parent fee request SMS alerts
Batch Standardization	GIET University B.Tech (2023 - 2027)	Uniform academic cohort schema and 4-year degree tracking